

HELSINKI VANTAA, Finland

WMO: 029740

Lat: 60.327N

Lon: 24.957E

Elev: 176

StdP: 14.60

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.7	-0.7	-11.6	3.0	-6.0	-5.6	4.1	-0.4	25.4	31.5	23.4	31.1	6.4	10	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	81.0	65.1	78.1	63.4	75.0	61.9	67.8	76.9	65.7	74.3	63.9	71.8	10.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.9	71.2	62.6	85.6	69.1	60.8	80.1	67.1	32.2	76.8	30.6	74.3	29.3	71.8	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.7	18.5	-12.2	84.9	7.6	3.3	-17.7	87.3	-22.1	89.2	-26.4	91.1	-32.0	93.4		
(5)			-12.6	70.4	7.5	1.8	-18.0	71.7	-22.4	72.8	-26.6	73.8	-32.1	75.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	23.2	22.4	28.7	40.1	50.7	58.8	64.5	61.8	53.0	42.4	34.2	28.0		
	DBStd	16.53	10.84	11.12	8.00	6.85	7.33	5.83	5.31	5.14	5.86	7.02	7.93	9.89		
	HDD50	4095	831	773	659	306	80	3	0	0	37	248	475	683		
	HDD65	8347	1296	1193	1124	748	447	203	75	127	360	701	925	1148		
	CDD50	1329	0	0	0	9	100	266	450	365	127	12	0	0		
	CDD65	106	0	0	0	0	3	16	60	26	1	0	0	0		
	CDH74	846	0	0	0	0	39	137	487	182	1	0	0	0		
	CDH80	132	0	0	0	0	3	13	96	20	0	0	0	0		
	WSAvg	9.1	9.7	9.4	9.3	8.8	9.0	8.9	8.5	8.1	8.6	9.4	9.8	10.2		
	PrecAvg	26.1	1.9	1.3	1.3	1.4	1.4	2.0	2.7	3.1	2.7	3.0	2.8	2.3		
(15) Precipitation	PrecMax	35.6	4.1	4.4	3.1	4.0	3.4	4.9	7.9	7.6	5.9	7.6	8.5	6.6		
	PrecMin	16.6	0.3	0.1	0.1	0.1	0.1	0.5	0.2	0.4	0.5	0.6	0.2	0.3		
	PrecStd	4.7	1.0	0.9	0.6	0.8	0.7	1.1	1.7	1.5	1.4	1.7	1.5	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	42.0	52.0	67.3	79.0	81.9	86.3	82.8	71.9	59.2	50.1	46.5		
		MCWB	40.1	37.9	43.6	51.9	60.9	63.8	67.2	66.0	60.2	53.5	48.5	45.3		
	2%	DB	38.9	38.0	46.0	61.5	73.5	77.9	82.6	78.7	68.1	56.1	48.2	44.0		
		MCWB	37.8	36.3	39.0	48.4	57.6	62.0	66.9	64.4	59.2	52.8	46.9	42.8		
	5%	DB	37.1	36.5	41.3	56.7	69.1	74.5	79.3	75.4	64.7	54.0	46.4	41.7		
		MCWB	36.3	35.3	36.5	46.3	55.1	60.6	64.7	62.4	57.6	51.6	45.2	40.6		
	10%	DB	35.4	35.2	37.8	52.2	64.5	70.9	76.2	72.2	62.4	52.1	44.6	39.4		
		MCWB	34.6	34.2	34.7	43.4	52.7	58.6	63.2	61.2	56.7	50.0	43.5	38.3		
	0.4%	WB	40.4	38.4	44.0	53.5	64.4	67.7	70.8	69.5	63.0	55.5	48.9	45.4		
		MCDB	40.8	39.9	51.1	64.8	76.3	77.6	80.0	77.9	67.6	57.3	49.6	46.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.9	36.7	39.7	49.9	60.1	64.8	68.8	66.6	61.1	54.0	47.0	42.8		
		MCDB	38.4	37.7	44.3	58.7	68.0	73.8	78.0	73.8	65.4	55.5	47.7	43.6		
	5%	WB	36.1	35.3	37.3	47.2	56.7	62.3	66.8	64.7	59.5	52.2	45.2	40.6		
		MCDB	36.7	36.1	40.4	55.0	66.6	71.0	76.3	71.6	63.1	53.8	45.9	41.5		
	10%	WB	34.5	34.0	35.3	44.5	53.8	60.0	64.9	63.1	57.8	50.5	43.5	38.4		
		MCDB	35.0	34.8	37.1	50.8	63.5	68.3	73.6	69.6	61.2	51.9	44.3	39.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	10.3	12.9	15.6	17.9	16.3	16.3	15.9	13.8	10.0	7.5	8.3		
		MCDBR	7.1	8.9	16.5	23.3	24.3	21.8	21.1	20.8	16.6	11.2	7.8	7.0		
		MCWBR	6.9	7.8	10.6	12.8	11.9	10.1	8.7	9.2	9.6	8.4	7.6	7.0		
		MCDBR	6.9	7.8	13.7	21.3	21.2	18.6	18.2	16.7	14.5	9.8	7.5	6.5		
	5% WB	MCWBR	6.8	7.1	9.6	12.4	11.3	9.7	8.7	8.7	9.3	8.2	7.5	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.310	0.332	0.358	0.356	0.354	0.375	0.368	0.341	0.331	0.302	N/A		
	taud		2.136	2.195	2.255	2.357	2.423	2.473	2.434	2.452	2.518	2.398	2.214	N/A		
	Ebn at Noon		150	213	250	266	276	279	268	261	250	208	139	N/A		
	Edh at Noon		13	23	30	32	32	31	32	29	23	19	13	N/A		
(44) All-Sky Solar Radiation	RadAvg		78	259	731	1218	1653	1717	1660	1282	782	342	96	46		
	RadStd		14	52	94	126	178	136	152	134	102	55	18	7		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.95	N/A	N/A	N/A	N/A	N/A	-261	-371	N/A	N/A
(47)	Regional (58 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	-259	N/A	N/A	N/A

Nomenclature: See separate page